

Aradhya Seed Store — Project Plan

Business requirements extracted from the handwritten layout, plus recommended free web and mobile technologies for development and deployment.

1. Business & Domain Information

Store Details

Field	Value
Business Name	ARADHYA SEED STORE
Location	Chhatter, India
Mobile	70180 63629

Purpose of the Application

Digitize two paper registers used daily at the store:

1. Stock / Inventory Register — track seed products currently in stock
 2. Customer Sales / Billing Register — record sales, payments, and outstanding balance
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2. Data Fields (from Handwritten Layout)

2.1 Stock Register (Inventory)

Used to track seeds in stock. Each row represents one product (or one batch).

Column	Description	Notes
Sr. No.	Serial number	Auto-generated row number
Product Name	Name of the seed product	e.g. Wheat, Mustard, etc.
HSN Code	Harmonized System of Nomenclature code	Required for GST invoicing in India
Quantity	Amount in stock	Units: Liters (Ltr.) or Kilograms (Kg.)
Manufacturing Date	Date product was manufactured	Track batch freshness
Expiry Date (Exp. Date)	Product expiry date	Important for seed quality & compliance

2.2 Customer Sales / Billing Register

Used when selling to customers. Header + line items + payment summary.

Customer Header

Field	Description
Customer Name & Address	Buyer details for invoice and records

Line Items (per product sold)

Column	Description	Notes
Sr. No. / Date	Entry number and transaction date	One invoice per sale
Product Name	Seed product sold	Linked to stock
HSN Code	Tax classification code	Same as on product
Quantity	Amount sold	Ltr. or Kg.
Rate	Price per unit	₹ per Kg or Ltr.
Amount	Line total	Quantity × Rate

Payment Summary (bottom of invoice)

Field	Description	Formula
Total Amount	Sum of all line amounts	Sum of Amount column
Received Amount	Cash/payment received from customer	Partial or full payment
Balance Amount	Outstanding amount owed	Total Amount – Received Amount

3. Functional Requirements (Derived)

Module	Features
Products & Stock	Add/edit products, HSN, unit (Kg/Ltr), MFG & expiry dates, current quantity
Sales / Invoices	Create bill with customer info, line items, auto-calculate totals
Customers	Store name, address, phone; track pending balance
Stock Deduction	Reduce stock automatically when a sale is saved
Reports	Stock list, sales by date, customers with outstanding balance, expiring stock
Print / Share	Printable invoice (A4 or thermal); optional WhatsApp share

Indian Market Considerations

- HSN codes on products and invoices (GST compliance)
- Units: Kilograms and Liters only
- Credit sales: Balance Amount implies simple accounts-receivable tracking
- Optional later: GST %, Hindi UI, UPI payment reference on invoice

4. Recommended Database Schema (Reference)

```
products      → id, name, hsn_code, unit, stock_qty, mfg_date, exp_date
customers     → id, name, address, phone
```

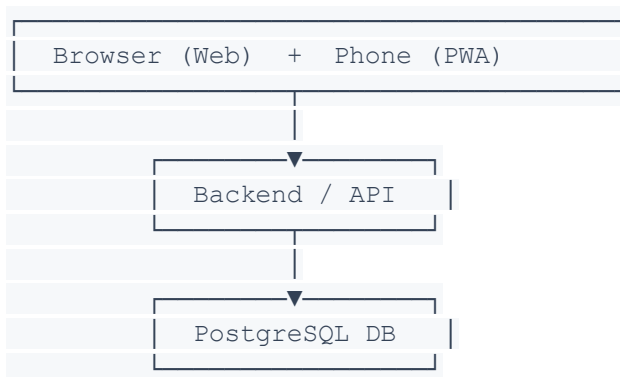
```
sales          → id, customer_id, date, total, received, balance
sale_items     → sale_id, product_id, quantity, rate, amount, hsn_code
stock_movements → product_id, type (in/out), quantity, reference (optional)
```

5. Technology Recommendations (Free Development & Deployment)

5.1 Strategy: One Codebase for Web + Mobile

Build a responsive web application that:

- Runs in any browser on desktop, tablet, and phone
- Can be installed on Android/iPhone like an app (PWA — Progressive Web App)
- Uses one backend and one database (no duplicate work)



5.2 Recommended Stack (Best Balance)

Layer	Technology	Why	Cost
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Frontend	Next.js (React)	Modern, fast, good for forms & tables	Free (MIT)
Styling	Tailwind CSS	Quick, clean UI	Free
Backend + DB	Supabase	PostgreSQL, auth, APIs, real-time	Free tier
Hosting (Web)	Vercel	Deploy Next.js in one click	Free tier
Mobile	PWA (Add to Home Screen)	No Play Store fee needed initially	Free
Invoices	jsPDF or <code>@react-pdf/renderer</code>	Generate PDF bills	Free
Version Control	GitHub	Store code, backups	Free

Total monthly cost for MVP: ₹0 (within free tiers)

5.3 Alternative Free Stacks

Approach	Stack	Best For
Simpler web	HTML + CSS + JavaScript + Supabase	Minimal learning curve
Python backend	FastAPI or Django + SQLite/PostgreSQL on Render	If you prefer Python

Firebase	React + Firebase (Firestore)	Quick start; less ideal for invoice relations
Native mobile later	Capacitor wrapping the web app	Free .apk build when you need a store app

5.4 Free Hosting & Deployment Options

Service	Use Case	Free Tier Highlights
Vercel	Host Next.js / React frontend	Generous bandwidth, auto HTTPS
Netlify	Alternative frontend host	Similar to Vercel
Supabase	Database + authentication + storage	500 MB DB, 50K monthly active users
Render	Host Node/Python API if not using Supabase-only	750 hrs/month (with limits)
Railway	Alternative backend host	Limited free credits
Cloudflare Pages	Static site / frontend	Unlimited bandwidth on free plan

Recommended deployment path: Next.js on Vercel + database on Supabase = no server management, ₹0 to start.

5.5 Free Domain Options

A custom domain (e.g. `aradhyaseedstore.in`) usually costs ₹500–800/year. These options are fully free:

Option	Example URL	Notes
Vercel subdomain	<code>aradhya-seed-store.vercel.app</code>	Instant, HTTPS, good for MVP
Netlify subdomain	<code>aradhya-seed-store.netlify.app</code>	Same idea
GitHub Pages	<code>username.github.io/aradhya-seed-store</code>	Free; static sites only
Cloudflare Pages	<code>aradhya-seed-store.pages.dev</code>	Free subdomain + CDN
Supabase project URL	For API only, not public marketing site	Backend use

Tip: Start with `*.vercel.app` or `*.pages.dev`. Buy a `.in` domain later when the business wants a professional URL (~₹99–800/year on promotions).

5.6 Mobile App Options (Free)

Method	Cost	Description
PWA (recommended first)	₹0	User opens site in Chrome → “Add to Home Screen” → icon on phone

Capacitor wrapper	₹0 dev	Same web app packaged as Android .apk for sideload or Play Store
React Native / Flutter	₹0 dev tools	Separate native app; more work; Play Store ₹1,750 one-time fee to publish

For a seed store in a town like Chhatter, PWA is sufficient for daily billing and stock entry on mobile.

6. Development Tools (All Free)

Tool	Purpose
VS Code or Cursor	Code editor
Node.js LTS	Run Next.js locally
Git	Version control
Chrome DevTools	Debug UI; test mobile layout
Figma (free tier)	Optional wireframes

7. Cost Summary

Item	Cost
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Development tools	Free
Hosting (Vercel + Supabase)	Free (free tiers)
Web URL (subdomain)	Free (*.vercel.app)
SSL / HTTPS	Free (included on Vercel/Netlify)
PWA mobile install	Free
Custom domain (optional)	~₹500–800/year
Google Play listing (optional)	₹1,750 one-time

Minimum to launch: ₹0

8. Suggested Build Phases

Phase 1 — Core MVP

- Product & stock CRUD (matches stock register columns)
- Create sale with customer, line items, total / received / balance
- Auto deduct stock on sale
- Owner login (Supabase Auth)

Phase 2 — Daily Use

- Customer list and outstanding balance report
- Invoice PDF + print

- Low stock and expiry alerts
- Search and filter products

Phase 3 — Mobile & Growth

- PWA offline support (weak network areas)
- Optional Android .apk via Capacitor
- Hindi labels (optional)
- GST % on invoice (if required)

9. Screen Layout (Mirror Paper Registers)

Stock Screen

Sr | Product Name | HSN | Qty | Unit | MFG Date | Exp Date | Edit

New Sale Screen

- Top: Customer name & address
- Middle: Line items — Product | HSN | Qty | Rate | Amount
- Bottom: Total Amount | Received Amount | Balance Amount
- Actions: Save | Print | Share

Dashboard (optional)

- Today's sales total
- Products expiring soon
- Total pending customer balance

10. Final Recommendation

Choice	Recommendation
Web framework	Next.js 14+ with App Router
Database & auth	Supabase (PostgreSQL)
CSS	Tailwind CSS
Hosting	Vercel
Free URL	<code>aradhya-seed-store.vercel.app</code> (or similar)
Mobile	PWA first; Capacitor later if needed
Learning order	HTML/CSS → JavaScript → React → Supabase

This stack keeps development and deployment free, scales with the business, and matches every field on the handwritten register.

Document created for Aradhya Seed Store, Chhatter — based on provided business layout.